



My-Woofie

A digital companion that looks after you while you work

Complete project guide: Version 1 and Version 2

Version 2.0.0

by Aashna Batabyal

github.com/aashnology/My-Woofie

1. My-Woofie at a glance

My-Woofie is a small pixel-art puppy who lives on top of your desktop. He wanders the screen edge, ambles after your cursor, and keeps quiet company while you work. After the focus time you choose (4 hours by default) he barks, shows a speech bubble with a specific suggestion, and counts down a break. When the break is over he calls you back with a soft chime.

He is a companion first and a timer second: small, calm, a little silly, and easy to read, change and run locally. Everything runs on your own computer, in plain Python with PyQt6.

The problem	Long stretches at a screen are easy to lose track of, and ordinary timers are easy to dismiss and forget.
The idea	A companion you actually like having around, who nudges you toward real breaks, and who is happier when you take them.
Who it is for	Anyone who works at a computer for hours: students, developers, writers, designers.
Privacy	No screen capture, no key logging, no file reading, no window titles, no network. Activity is just elapsed time and mouse movement.
Platforms	Windows and macOS (Python 3.9+), with packaged downloads; also runs from source on Linux.

Number	What it counts
6	pixel-art dogs (Aspen, Biscuit, Cocoa, Bailey, Benny, Snow)
53 x 47	pixels: the exact on-screen size of every dog
4 h / 15 min	default focus time and break length (both adjustable)
13	headline features added in Version 2
101	automated tests (19 in Version 1)
0	network calls, screenshots, keystrokes or window titles recorded

2. Meet the pups

Six dogs are part of My-Woofie. Pick yours on first launch, and change your mind at any time from the right-click menu. Aspen, Biscuit and Cocoa are drawn in code from a palette. Bailey, Benny and Snow are built from the original design sheet. Every dog has ten animation frames: idle, walking, barking, the back-to-work hop and a happy pose.



Aspen
Golden retriever



Biscuit
Cream puppy



Cocoa
Chocolate lab



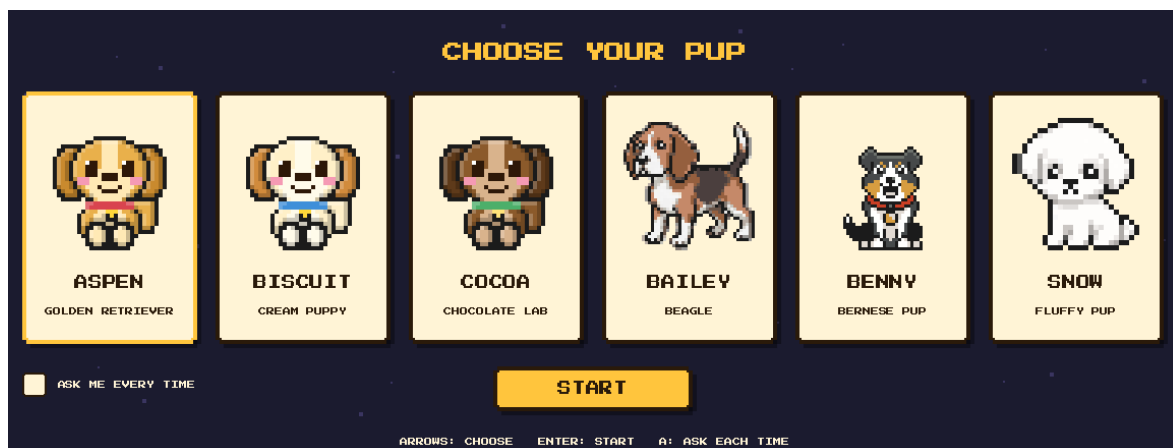
Bailey
Beagle



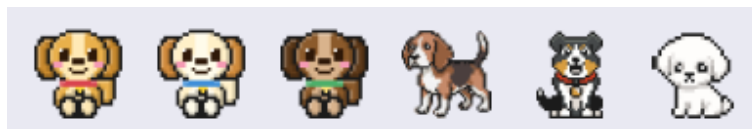
Benny
Bernese pup



Snow
Fluffy white pup



The retro avatar picker, with the "ask me every time" option.



Every dog is exactly 53 x 47 pixels on screen, shown here at true size.

3. Version 1: the foundation

Version 1 delivered the complete core experience: a calm companion, a reliable timer, and a retro look.

Feature	What it does
Six pixel-art dogs	Fixed 53 x 47 pixel canvas, ten frames each, picked in a retro avatar selector.
Avatar picker	Choose on first launch; tick "ask me every time" to choose at every start. Dismissing is never remembered.
Cursor following	Slow, calm movement. He trots toward the cursor while you move the mouse and roams the screen edge when you are still.
Focus and break timer	4 hours of focus, 15-minute break by default, adjustable by dialog, command line or config.
Presence detection	Only 15 continuous minutes without mouse movement resets the focus timer, which means you were genuinely away.
Sound	Bark and soft howl on break start, soft chime on break end, synthesized in code, with system-beep fallbacks if audio fails.
Speech bubbles	Pixel bubbles with rotating messages from break_prompts.txt and focus_prompts.txt. Always shown in silent mode.
Retro HUD	Focus meter that turns green, amber, red; break countdown; flashing back-to-work banner.
Menu and tray	Pause, timer settings, change avatar, focus meter, mute, reset, quit.
Easy to stop	Quit from the menu, or python main.py --stop from any terminal. Single-instance protection.
Local only	Settings saved in a local JSON file. No network, no screen capture, no key logging.



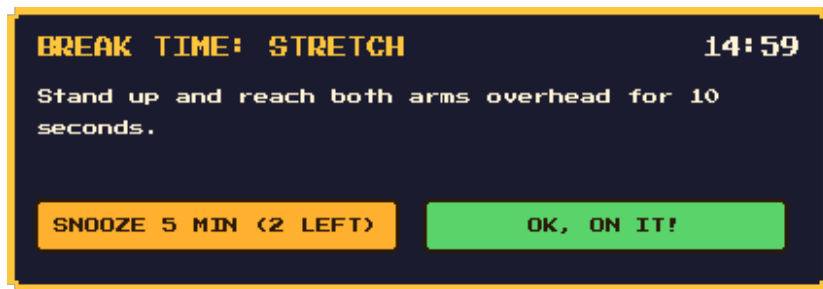
The retro HUD and a speech bubble during a break.

4. Version 2: what is new

Version 2 turns the timer into a companion with habits, moods and things to do, while keeping the same privacy rules. Every feature below is built, tested and merged.

Feature	What it does
1. Snooze with a cost	"Snooze 5 min" on the break card, limited per focus cycle (default 2, adjustable 0 to 5). Each snooze lowers his mood and is counted in the stats.
2. Smarter breaks	Short micro-breaks every 20 minutes: a 20-second eye rest, stretch, water sip or posture check, with Done and Skip. They never fire right before a big break, while you are away, or in fullscreen.
3. Break guidance	The break card suggests something specific (stretch, water, a short walk, breathing), rotating every 25 seconds from <code>break_tips.txt</code> .
4. Settings window	Right-click, Settings: five tabs (Timers, Pet, Sound, System, Data) with timers, speed, volume, night behaviour, start at login and data controls.
5. Mood and streaks	Real breaks make him happy; skipped breaks and snoozes make him droopy and slower. Healthy days build a streak (one missing day is forgiven).
6. Petting, treats, fetch	Click and hold to pet him (hearts). Give a treat (5 a day). Play fetch: drag and throw the ball, he chases it and brings it back to your cursor.
7. Time of day	Curls up for a nap when you are idle at night, stretches and greets you in the morning, and gives a gentle bedtime nudge when it is late.
8. Wardrobe	Party hat, flower, beanie, bandana, top hat and crown, unlocked by streak days (1, 3, 5, 7, 10, 14).
9. Fullscreen awareness	While a program covers a whole monitor, or in Presentation mode, he hides and alerts wait. An overdue break fires as soon as it ends.
10. Multi-monitor	He wanders between monitors, chases the cursor across them, and shows alerts on the monitor your cursor is on.
11. Start at login and tray	Optional autostart on Windows, macOS and Linux. Click the tray icon for status, double-click to pause.
12. Local-only stats	Weekly summary of breaks taken and skipped, snoozes, focus time and streak. One-click Delete my data, <code>--stats</code> and <code>--delete-data</code> .
13. Packaged builds	PyInstaller recipe and a GitHub Actions workflow that builds a Windows <code>.exe</code> and a macOS <code>.app</code> and attaches them to a Release.

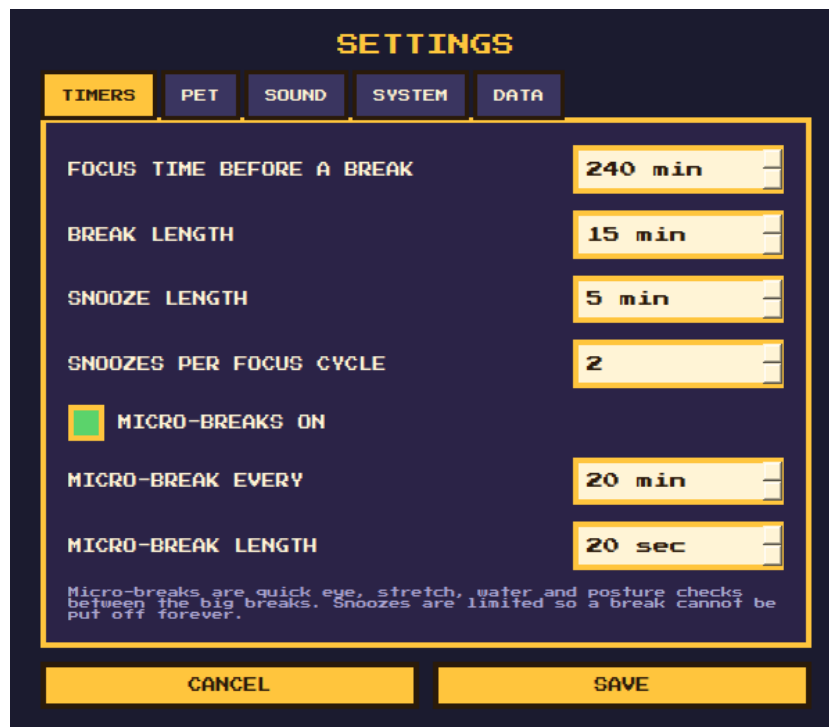
The new screens



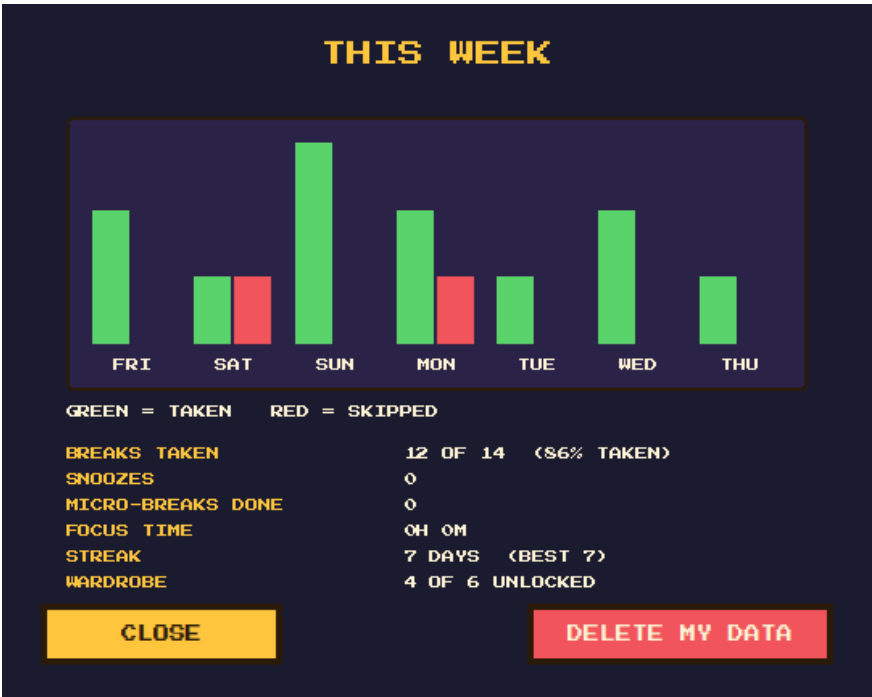
The break card: a specific suggestion, a countdown, Snooze (with snoozes left) and OK.



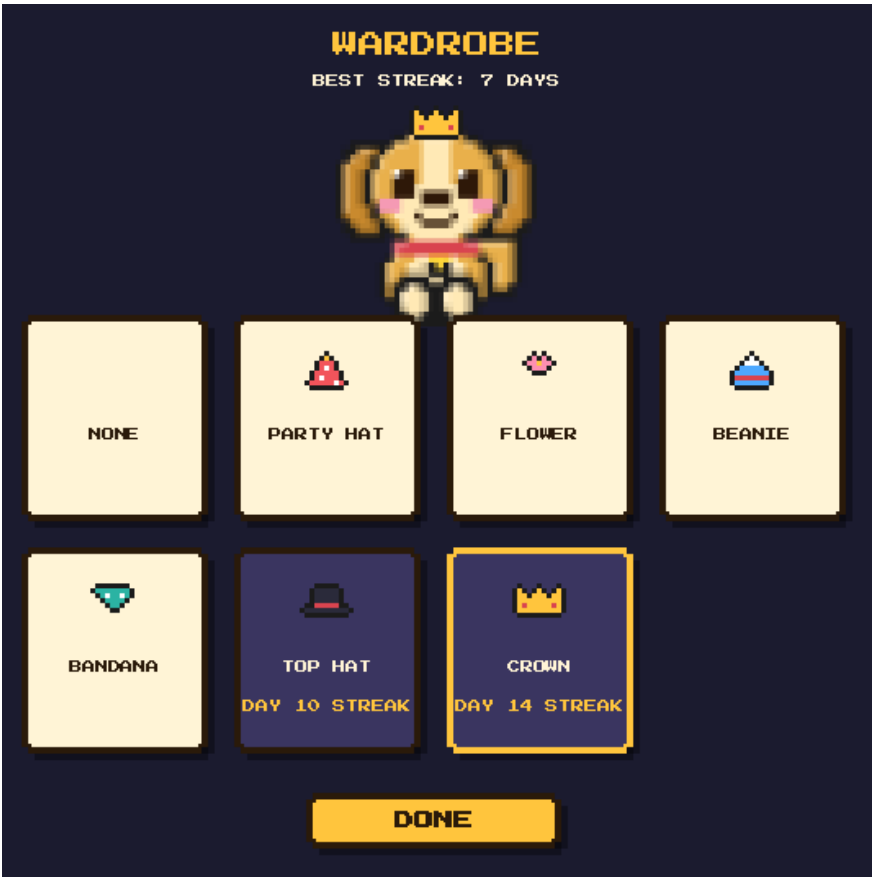
A micro-break card between big breaks: Done or Skip.



The settings window. The same options are saved to settings.json.



The weekly summary: green bars are breaks taken, red are skipped. Stored only on your computer.



The wardrobe. Locked items show the streak they need.

Accessories on every dog

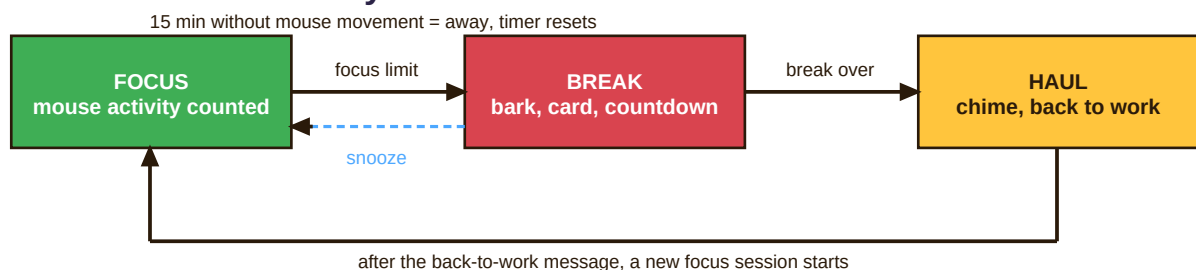
Each accessory is drawn onto the same 53 x 47 canvas, so the window never changes size. The dog is drawn a few pixels smaller to make room for hats. Rows: party hat, flower, beanie, top hat, crown, bandana.



5. A day with My-Woofie

What you are doing	What he does
Moving the mouse	Ambles toward your cursor and sits near it (on whichever monitor you are using).
Keeping the mouse still	Wanders slowly along the screen edge and rests now and then.
Clicking and holding on him	Petting: happy hop, hearts, his mood goes up.
Every 20 minutes of focus	A quick eye rest, stretch, water or posture card appears. Click Done or Skip.
Working past your focus time	Walks to mid-screen, barks, shows a break card with a tip and a Snooze button, and counts down your break.
You step away during the break	The break is recorded as taken, his mood rises, and the streak grows.
You keep working through it	The break is recorded as skipped and his mood drops.
Break time is over	Soft chime, a back-to-work bubble, then back to wandering.
Idle at night	Curls up and naps with Zzz; wakes the moment you move the mouse.
First activity in the morning	A stretch and a greeting.
Late at night, still working	A gentle bedtime nudge, repeated every hour.
A presentation or fullscreen app	Hides and holds alerts until it ends.
You right-click him	Menu: pause, snooze, presentation mode, treat, fetch, wardrobe, weekly summary, settings.

The focus and break cycle



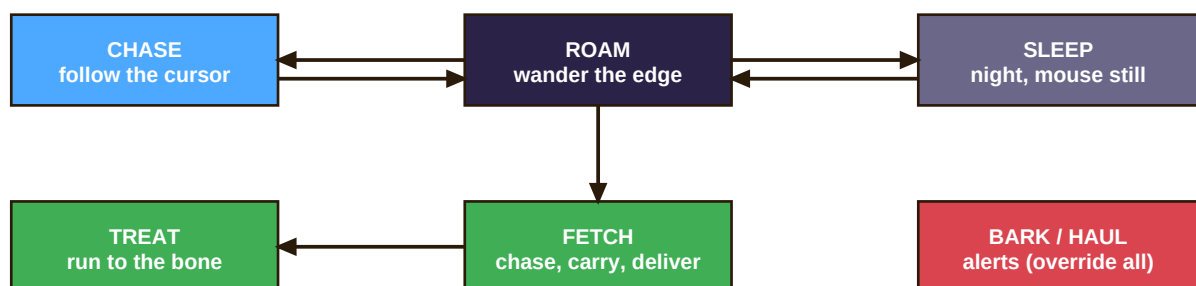
A break counts as **taken** if the mouse was active for no more than 25% of it, and as **skipped** otherwise. Only mouse movement is used, never what is on screen.

6. How it works

The code is split into small, readable modules so every feature can be debugged and changed on its own in VS Code. The timer, pet, stats, mood and scheduling logic have no Qt dependency, so they are tested with a fake clock.

Module	Role
main.py	Thin launcher: command-line options, single-instance check, startup dialogs.
companion.py	WoofieApp: owns every window and part and runs the 33 ms main loop.
timer.py	Session timer: presence detection, FOCUS / BREAK / HAUL, snooze, break quality, deferral.
pet.py	Finite state machine and movement: roam, chase, sleep, treat, fetch, alerts, multi-monitor.
gui.py, retro.py	Transparent click-through windows, sprites, HUD, bubbles, emotes, toys, break card; retro pixel styling.
dialogs.py	Settings window, weekly summary, wardrobe.
stats.py, mood.py	Per-day counters and streaks; mood, daily limits, wardrobe state.
microbreak.py, tips.py, daypart.py	Micro-break scheduler, tip bank, night / morning / bedtime rules.
toys.py, wardrobe.py	Ball physics and bone; pixel-art accessories drawn onto frames.
fullscreen.py, autostart.py, paths.py	Fullscreen detection (sizes only), start at login, source vs packaged file locations.
assets_builder.py, audio.py	Sprite generation for all dogs; synthesized sounds with beep fallbacks.
settings.py, prompts.py, selector.py, timer_dialog.py, instance.py	Saved settings, message banks, avatar picker, first-run timer dialog, single instance and --stop.

The pet's state machine



A break or back-to-work alert interrupts every other state, then he returns to ROAM

Main loop. Every 33 ms the app reads the cursor position, polls fullscreen state every 3 seconds, feeds the timer, turns timer events into statistics and mood changes, updates the micro-break scheduler and the day rhythm, moves the pet, moves toys and then redraws windows. The timer takes an injectable clock, so hours of behaviour are simulated instantly in tests.

Click-through windows. The dog, bubble, emotes and HUD ignore the mouse, so clicks pass to the apps underneath. Only the dog's own pixels (a mask), the fetch ball and the break card accept clicks, and they never steal keyboard focus.

7. Privacy and safety by design

These rules apply to both versions and are enforced by how the code is written, not by a setting.

My-Woofie never...	How it is guaranteed
captures the screen	No screen-capture API is imported or called anywhere.
logs keystrokes	Only the cursor position is read, once per tick. There is no keyboard hook.
reads files or window titles	No code reads other programs' files, titles or names.
uses the network	No sockets to the internet. The only socket is a local one used for --stop and single-instance checks.
detects fullscreen by looking at your screen	It compares the foreground window's size with the monitor size. On macOS it reads only window layer, transparency and bounds.
sends your statistics anywhere	stats.json holds integer counters per date and stays in the project or user folder. Delete my data removes it.
locks or blocks you	He nudges; you decide. Snooze, Skip and Pause are always available (snooze is limited per cycle by design).

A limit you should know: presence comes only from time and mouse movement, so reading or typing without touching the mouse for 15 minutes counts as being away.

8. Install, run and options

Requirements: Python 3.9 or newer on Windows or macOS.

```
git clone https://github.com/aashnology/My-Woofie.git
cd My-Woofie
python -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
pip install -r requirements.txt
python main.py
```

On first launch you choose a pup and your focus and break times. After that he starts straight away.

I want to...	Do this
Stop him	Right-click, Quit My-Woofie. Or <code>python main.py --stop</code> in any terminal.
Choose a different dog	Right-click, Change avatar... (or <code>python main.py --select</code>).
Start over	Right-click, Reset all settings... (or <code>python main.py --reset</code>).
Change timers, speed, sound, night behaviour	Right-click, Settings...
Snooze a break	Click SNOOZE on the break card, or right-click, Snooze this break.
Hold alerts for a presentation	Right-click, Presentation mode.
See my week	Right-click, Weekly summary... (or <code>python main.py --stats</code>).
Delete my statistics	Settings, Data, Delete my data (or <code>python main.py --delete-data</code>).
See a full cycle quickly	<code>python main.py --test</code> (10 s focus, 5 s break; keep moving the mouse).

All options: `--select`, `--reset`, `--settings`, `--stop`, `--test`, `--focus MIN`, `--break MIN`, `--stats`, `--delete-data`, `--version`.

Customizing

- **Messages:** one per line in `break_prompts.txt` and `focus_prompts.txt` (# starts a comment). Changes apply without restarting.
- **Break tips:** `break_tips.txt`, one per line as category|text (eyes, stretch, water, posture, walk, breathe).
- **Sounds:** replace the files in `assets/audio/`, or change `SOUND_EVENTS` in `config.py`.
- **Everything else:** constants at the top of `config.py`, and the defaults in `settings.py`.

Settings reference

Setting	Default	Meaning
Focus time / break length	240 min / 15 min	When he asks for a break, and how long it lasts.
Snooze length / snoozes per cycle	5 min / 2	How far a snooze postpones the break, and how many you get.
Micro-breaks	on, every 20 min, 20 s	Short eye / stretch / water / posture reminders.
Walking speed	3 of 5	Scales how fast he moves.
Wander between monitors	on	Whether he visits your other screens.
Night naps and bedtime nudge	on; night 22:00 to 06:00; nudge from 23:00	Time-of-day behaviour.

Setting	Default	Meaning
Volume / mute	80% / off	Sound level, or speech bubbles only.
Hold alerts during fullscreen	on	Fullscreen and presentation awareness.
Always show focus meter	off	HUD only during breaks by default.
Keep break statistics	on	Local-only weekly summary and streaks.
Start at login	off	Autostart on Windows, macOS and Linux.

9. Packaging, testing and quality

Building a download

```
pip install -r requirements-dev.txt
python packaging/build.py      # runs the tests, then PyInstaller; result in dist/
```

Build on the system you are targeting: Windows gives a .exe, macOS gives a .app. Pushing a tag such as v2.0.0 runs .github/workflows/release.yml, which tests the code and builds both versions on GitHub, then attaches them to a Release. Packaged builds keep settings, stats and the editable text files in your user folder instead of the program folder.

Tests

```
python -m unittest discover -s tests
```

101 automated tests run in about a second with no real screen needed. They cover the timer (cycle, limits, away reset, jitter, snooze, break quality, deferral), the pet (states, chasing, sleep, treats, fetch, multi-monitor, bounds), stats and streaks, mood and daily limits, micro-breaks, day rhythm, tips, ball physics, start at login, fullscreen geometry, the settings window, and that every dog stays exactly 53 x 47 pixels with every accessory.

What was verified, and what was not

Verified here	Not yet verified on real hardware
All 101 tests; headless full-cycle runs (treat, fetch, micro-break, break, haul); visual checks of every accessory on every dog; a PyInstaller build and start on Linux.	Real Windows and macOS desktops (click-through masks, tray notifications, audio devices), real multiple monitors, real fullscreen apps, and the Windows / macOS CI builds, which are unsigned so the OS may show a security prompt on first launch.

10. Known limitations and ideas for next

Known limitations

- Presence is mouse-only by design, so working without touching the mouse looks like being away.
- Automatic fullscreen detection works on Windows; it is off on Linux and needs the optional Quartz package on macOS. Presentation mode works everywhere.
- A micro-break that times out on its own is not counted as done; only a click on Done is recorded.
- The sleep pose reuses his happy frame, so he naps with his tongue out.
- The default sounds are synthesized and sound artificial compared with real recordings.
- Window masks for click-through can behave differently on macOS.

Ideas for a future version (not built)

- Real recorded bark and chime sounds.
- More dogs, and a way to add your own from a PNG sprite sheet.
- Optional keyboard-activity awareness that never records what is typed.
- Per-day goals and gentle weekly reports.
- Signed installers for Windows and macOS.

11. Hackathon quick reference

The 60-second demo

```
python main.py --test
```

Time	What to show
0 s	Pick a dog in the retro picker. Move the mouse and watch him follow the cursor.
6 s	A micro-break card appears (eye rest). Click Done: mood goes up.
10 s	The break starts: he barks, the card shows a tip and a Snooze button, the HUD counts down.
Click Snooze	The break moves away for a few seconds, then returns. Snoozes are limited.
During the break	Hold the mouse still to take the break, or keep moving to skip it.
After the break	Soft chime and back-to-work banner. Open Weekly summary to see taken versus skipped.
Right-click	Give a treat, play fetch (throw the ball), open the wardrobe, open Settings.

The pitch in three sentences

Breaks only work if you actually take them, and a plain timer is easy to ignore. My-Woofie is a pixel puppy who stays on your desktop, nudges you toward real breaks with specific suggestions, and gets happier when you take them. It does all of this locally, with no screen capture, no key logging and no network, so it is safe to leave running all day.

License and links

Released under the MIT License. Copyright 2026 Aashna Batabyal. Source: github.com/aashnology/My-Woofie. The repository also contains the User Guide and Technical Guide (docs/), the changelog, and the unit tests.